NOTENEST | MODEL PRACTICE PAPER | DEMO

# Computer Networks Model Practice Paper

Computer Networks - Demo paper - 2023 edition - 50 marks

> *Self-created practice paper for NoteNest. Not an official university paper.*

## Section A - Short answers (4 x 5 marks)

**Q1.** Compare the OSI and TCP/IP reference models. **[5 marks]**

**Q2.** How many usable hosts does a /27 network provide? Write its subnet mask. **[5 marks]**

**Q3.** Differentiate between TCP and UDP and give one application of each. **[5 marks]**

**Q4.** Explain the role of DNS. **[5 marks]**

## Section B - Long answers (2 x 15 marks)

**Q5.** Explain the TCP three-way handshake and the slow-start / congestion-avoidance phases of TCP congestion control with a diagram. **[15 marks]**

**Q6.** Compare distance-vector and link-state routing. Describe the count-to-infinity problem and one remedy. **[15 marks]**

## Answer hints

- Q2: a /27 leaves 5 host bits, so 2^5 - 2 = 30 usable hosts; mask 255.255.255.224.
- Q6: remedies include split horizon and route poisoning.